

4.3 Maximum & Minimum Values Worksheet

1. Find the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 + 1$ on the interval $\left[-\frac{1}{2}, 4\right]$.

Since f is continuous on $\left[-\frac{1}{2}, 4\right]$ it must have an absolute min. & max. on the interval $\left[-\frac{1}{2}, 4\right]$.

We calculate: $f'(x) = 3x^2 - 6x$.

Notice that $f'(x)$ is never undefined.

Notice that $f'(x) = 0$ when $0 = 3x^2 - 6x \rightarrow 0 = 3x(x - 2) \rightarrow x = 0$ or $x = 2$

Therefore, the critical numbers of the function $f(x)$ are $x = 0$ & $x = 2$.

We also consider the endpoints of the given interval $x = -\frac{1}{2}$ & $x = 4$.

$$f(0) = 1 \quad \& \quad f(2) = -3 \quad \& \quad f\left(-\frac{1}{2}\right) = \frac{1}{8} \quad \& \quad f(4) = 17$$

Therefore, we see that $f(x)$ has:

- an absolute minimum $f(2) = -3$
- an absolute maximum $f(4) = 17$

2. Find the absolute maximum and minimum values of $f(x) = x^3 - 12x + 1$ on the interval $[0, 5]$.

Since f is continuous on $[0, 5]$ it must have an absolute min. & max. on the interval $[0, 5]$.

We calculate: $f'(x) = 3x^2 - 12 = 3(x^2 - 4)$

Notice that $f'(x)$ is never undefined.

Notice that $f'(x) = 0$ when $x = \pm 2$. Since -2 is not in the interval $[0, 5]$ we only consider $x = 2$.

The values of f at our critical numbers and endpoints are:

$$f(0) = 1 \quad \& \quad f(2) = -15 \quad \& \quad f(5) = 66$$

Therefore, we see that $f(x)$ has:

- an absolute minimum $f(2) = -15$
- an absolute maximum $f(5) = 66$

3. Find the absolute maximum and minimum values of the function $f(x) = (x^2 + x)^{2/3}$ on the interval $[-2, 3]$.

Since f is continuous on $[-2, 3]$ it must have an absolute min. & max. on the interval $[-2, 3]$.

We calculate: $f'(x) = \frac{2}{3}(x^2 + x)^{-1/3}(2x + 1)$

Notice that $f'(x)$ is undefined when $x^2 + x = 0 \rightarrow x(x + 1) = 0 \rightarrow x = 0$ or $x = -1$

Notice that $f'(x) = 0$ when $2x + 1 = 0 \rightarrow x = -\frac{1}{2}$

Therefore, the critical numbers of the function $f(x)$ are $x = -\frac{1}{2}$, $x = -1$, & $x = 0$.

We also consider the endpoints of the given interval $x = -2$ & $x = 3$.

The values of f at our critical numbers and endpoints are:

$$f(-2) = \sqrt[3]{4} \quad \& \quad f\left(-\frac{1}{2}\right) = (-0.5)^{2/3} \quad \& \quad f(-1) = 0 \quad \& \quad f(0) = 0 \quad \& \quad f(3) = \sqrt[3]{144}$$

Therefore we see that $f(x)$ has:

- an absolute minimum $f(-1) = f(0) = 0$
- an absolute maximum $f(3) = \sqrt[3]{144}$

4. Find the absolute maximum and absolute minimum values of $f(x) = -2x\sqrt{x^2 + 9}$ on the interval $[-4, 1]$.

Since f is continuous on $[-4, 1]$ it must have an absolute min. & max. on the interval $[-4, 1]$.

We calculate:

$$f'(x) = -2\sqrt{x^2 + 9} - x(x^2 + 9)^{-1/2}(2x) = -2(x^2 + 9)^{-1/2}(x^2 + 9 + x^2) = -2(x^2 + 9)^{-1/2}(2x^2 + 9)$$

Notice that $f'(x)$ is never undefined.

Notice that $f'(x)$ never equals 0.

Therefore, we must only check the endpoints of the interval.

$$f(-4) = 40 \quad \& \quad f(1) = -2\sqrt{10}$$

Therefore, we see that $f(x)$ has:

- an absolute minimum $f(1) = -2\sqrt{10}$
- an absolute maximum $f(-4) = 40$

5. Find the absolute maximum and minimum values of $g(t) = t + \cot\left(\frac{t}{2}\right)$ on the interval $\left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$.

Since g is continuous on $\left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$ it must have an absolute min. & max. on the interval $\left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$.

We calculate: $g'(x) = 1 - \frac{1}{2} \csc^2\left(\frac{t}{2}\right)$

Notice that $g'(t)$ is undefined if $\frac{t}{2} = \pi \rightarrow t = 2\pi$. Since none of these values are within the given interval, $g'(t)$ is never undefined.

Notice that $g'(t) = 0$ when:

$$0 = 1 - \frac{1}{2} \csc^2\left(\frac{t}{2}\right) \rightarrow 2 = \csc^2\left(\frac{t}{2}\right) \rightarrow \pm\sqrt{2} = \csc\left(\frac{t}{2}\right) \rightarrow \pm\frac{\sqrt{2}}{2} = \sin\left(\frac{t}{2}\right).$$

This implies that $t = \frac{\pi}{2}$ & $\frac{3\pi}{2}$

Therefore, the critical numbers of the function $g(t)$ are: $t = \frac{\pi}{2}$ & $\frac{3\pi}{2}$.

We also consider the endpoints of the given interval $t = \frac{\pi}{4}$ & $\frac{7\pi}{4}$.

The values of g at our critical numbers and endpoints are:

$$g\left(\frac{\pi}{4}\right) = \frac{\pi}{4} + \cot\left(\frac{\pi}{8}\right) \approx 3.200 \quad \& \quad g\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + 1 \approx 2.571 \quad \& \\ g\left(\frac{3\pi}{2}\right) = \frac{3\pi}{2} - 1 \approx 3.712 \quad \& \quad g\left(\frac{7\pi}{4}\right) = \frac{7\pi}{4} + \cot\left(\frac{\pi}{8}\right) \approx 3.084$$

Therefore, we see that $g(t)$ has:

- an absolute minimum $g\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + 1 \approx 2.571$
- an absolute maximum $g\left(\frac{3\pi}{2}\right) = \frac{3\pi}{2} - 1 \approx 3.712$